QMSS 301: Quantitative Social Science Analysis and Big Data

Lab Assignment – Python Modules

Due on October 4, 2023, 11:59pm

We will be working with a new dataset about Pokémon information for this assignment. Please download the ***'pokemon.csv'*** file along with the assignment.

1. a. Install the matplotlib package
   b. Import matplotlib and pandas packages
   c. Create a dataFrame *'poke_data'* from the pokemon csv file by reading it using the pandas module. Print out the dataset and use a function that would give you the number of rows and columns for this dataset. Write down your observations.
   [Refer Numpy_and_Pandas.ipynb - Q1]

2. a. The first column of the DataFrame is not the name of a Pokemon but its abilities. We prefer to set the first column to be the name of a Pokemon, for better readability.
   Hint: Use *.pop()* to remove *'name'* column and then *.insert()* to add *'name'* to desired index location [Refer - Introduction_to_Python.ipynb]

   b. Once you make this change, print out only the first 9 rows of the dataset to verify your column change.
   Optional: Pokemon knowledge check! What seems interesting to you about the first 9 rows?

3. a. Look at the summary statistics on the Base Stats of the Pokemon. The Base Stats of consists of the *['hp', 'attack', 'defense', 'sp_attack', 'sp_defense', 'speed']*. Store your statistics in a new object called *'poke_base_stats'*. **[Refer Numpy_and_Pandas.ipynb - Q3]**

   b. Do the same for the other numerical data, which include the [*'height_m'*, *'weight_kg', 'capture_rate', 'base_egg_steps', 'experience_growth'*, *'base_happiness'*]. Store your statistics in a new object called *'poke_other_stats'*.

   Make two interpretations (one from each) of your choice from the two new statistics created.

4. Now, we will explore the data by creating different charts and plots. We begin with a simple one. Create a bar chart to show the number of new Pokemon in each *'generation'*. Give a suitable plot title and x,y labels. **[Refer Matplolib.ipynb]**
   Hint: Use .value_counts() function to give you number of pokemon by generation - Refer Numpy_and_Pandas.ipynb - Q9

5. a. Create a scatter plot to see how *'attack'* pokemons fare against *'defense'* pokemons. Give a suitable plot title and x,y labels. **[Refer Matplolib.ipynb]**

   b. Filter those pokemons with an attack score of greater than/equal to 140 and a defense score of greater than/equal to 130 and store this in a new variable *'high_attack_defense'*.
   [Refer Numpy_and_Pandas.ipynb - Q7]

   c. Define a new function called *'legendary_check(x)'*. The function should return **True** if a pokemon has *'is_legendary'* = 1 and **False** if *'is_legendary'* = 0. Using the *'high_attack_defense'* data you created, filter out *'Groudon'* and *'Salamence'* from the *'name'* column and check with your function whether they are legendary pokemon or not.
   [Refer Numpy_and_Pandas.ipynb - Q11]

6. Create a histogram to show the frequency distribution of the pokemon weight in the column *'weight_kg'*. Use bins= 50, label your y-axis as 'Count of Pokemon' and x-axis as 'Weight in Kgs'. Give a suitable plot title. **[Refer Matplolib.ipynb]**

7. Create a multiple lines graph showing the mean values of the [*'hp', 'attack', 'defense', 'speed'*] stats when grouped by *'generation'*. Give a suitable plot title and x,y labels.
   [Refer Matplolib.ipynb]
   Use the following formatting:
   'b--' blue dashed line for ['hp']
   'rs-' red squares with line for ['attack']
   'g^-' green triangles with line for ['defense']
   'md-' magenta diamonds with line for ['speed']

8. Similar to how we created a histogram for generations in question 4, create two histograms counting the number of pokemon by types.
   Histogram 1 - count by *'type1'*
   Histogram 2 - count by '*type2'*
   Create these histograms as subplots within one figure. First, create a figure size of *(15,10)* units.
   Add both the histogram subplots as - *fig.add_subplot(2,1, histogram position)*
   Rotate x-axis labels for better readability (optional) after each *plt.bar()* code by adding this code - *plt.xticks(rotation=45)*
   [Refer Matplolib.ipynb]

Please submit your .ipynb format file via Canvas by the deadline. Please use the naming convention **'Lab3_YourName'**. Example – **Lab3_VardhJain**.